A

BRAC UNIVERSITY
Department of Computer Science & Engineering
FINAL EXAMINATION, Summer 2025
CSE 350: Digital Electronics and Pulse Techniques

Total Marks: 50
Time: **2 Hours**

- There are **four(4)** questions. Answer them **all**. Question **4** is a **bonus** question.
- Figure in bracket [ ] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet.
- $V_{BE}(saturation) = 0.8\,V$, $V_{CE}(saturation) = 0.2\,V$, $V_{BC}(reverse\,active) = 0.7\,V$, $V_D(forward\,bias) = 0.7\,V$ if not stated otherwise.

Name:	ID:	Section:

Question 1: **[20]**

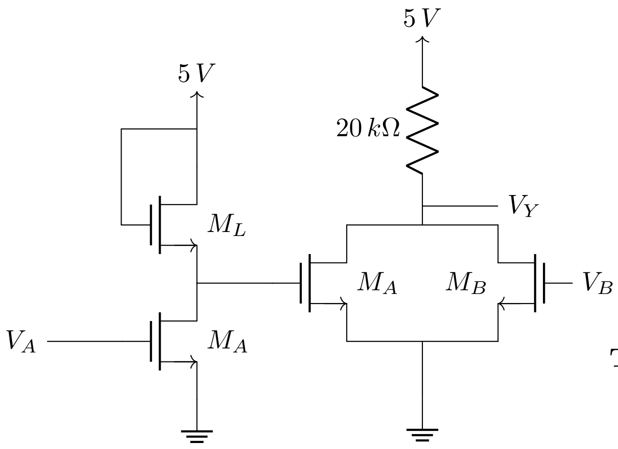

Figure 1

	$K_n\left(\frac{mA}{V^2}\right)$	V_{TN} (V)
M_L	0.3	1.5
M_A	0.75	0.6
M_B	0.65	0.5

Table 1: MOSFET parameters

Triode Mode: $I_{DS} = K_n[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$

Saturation Mode: $I_{DS} = K_n(V_{GS} - V_{TN})^2$

CO1	(a)	Write the **Boolean function** expression of the output (Y) in terms of the inputs (A & B).	[2]
CO1	(b)	Find the **power dissipation** of the **full circuit** when $V_A = V_B = 5$ V (high).	[5]
CO1	(c)	If the $20\,k\Omega$ was replaced with M_L, find the **output voltage** V_Y when, $V_A = 0$ V (low) & $V_B = 5$ V (high).	[5]

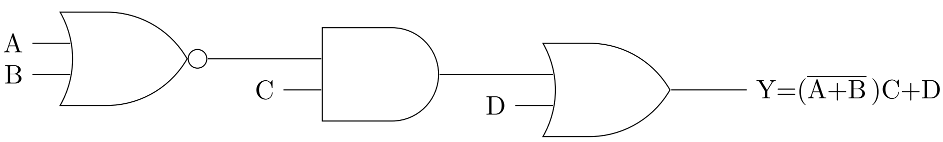

Figure 2

CO1	(d)	Implement the Boolean function shown in Figure 2 using **CMOS logic**.	[8]

Question 2: **[20]**

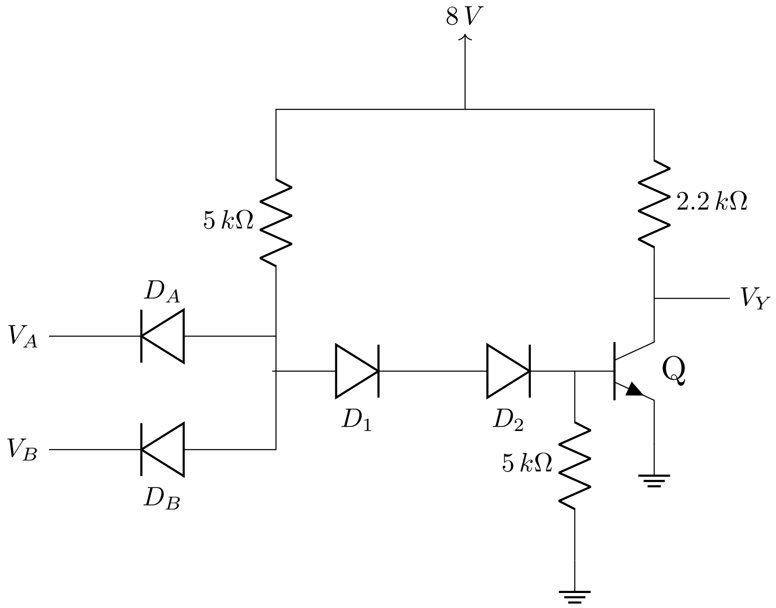

Figure 3

For the circuit shown in Figure 3, $\beta_F = 25$, $V_{OL} = 0.2\,V$, high input = 8 V, low input = 0.2 V.

CO2	(a)	Find the **average power dissipation** of the circuit.	[5]
CO2	(b)	If the circuit in Figure 3 is used as both driver & load, find the **maximum fanout**.	[5]

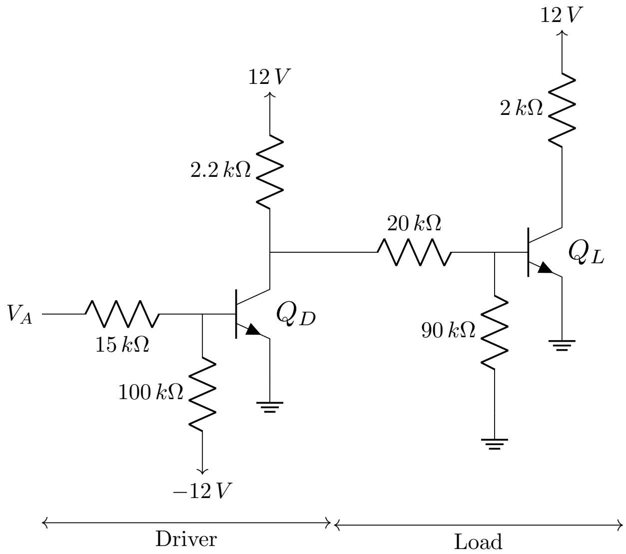

Figure 4

For the circuit shown in Figure 4, $\beta_F = 25$, $V_{OH} = 10\,V$, $V_{OL} = 0.2\,V$, high input = 12 V, low input = 0.2 V.

CO2	(c)	Find the **noise margin** of the circuit.	[5]
CO2	(d)	Find the **maximum fanout** of the driver.	[5]

Question 3: **[10]**

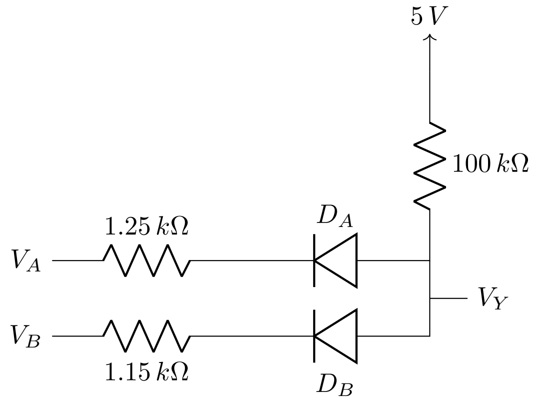

Figure 5

CO1	(a)	For the circuit in Figure 5, find the **output voltage** V_Y and the **power dissipated** when V_A = high & V_B = low. [high input = 5 V, low input = 0 V]	[5]

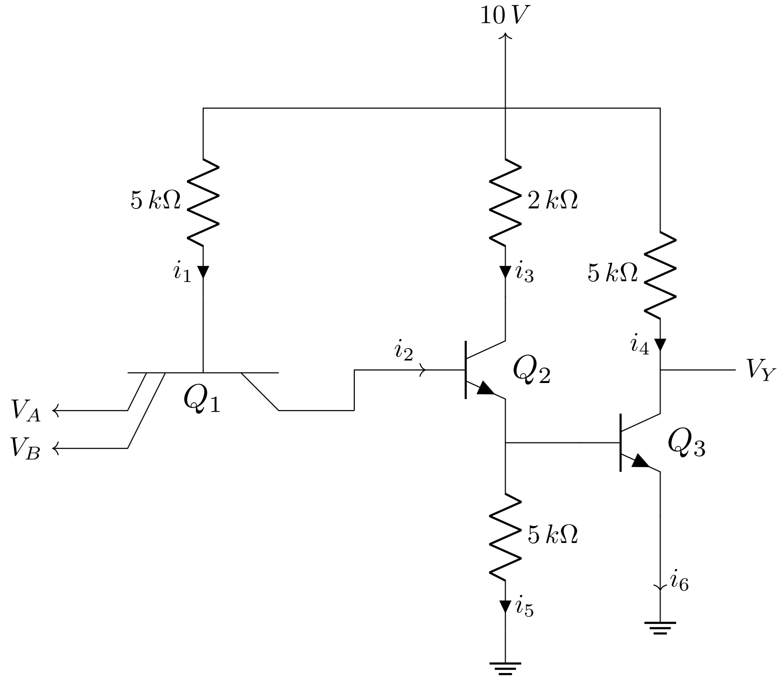

Figure 6

For the circuit shown in Figure 6, high input = 10 V, low input = 0.2 V.

CO1	(b)	If $\boldsymbol{i_6 = 8.302\,mA}$, find the value of **reverse beta**, $\boldsymbol{\beta_R}$ of Q_1.	[5]

Question 4: [Bonus] **[10]**

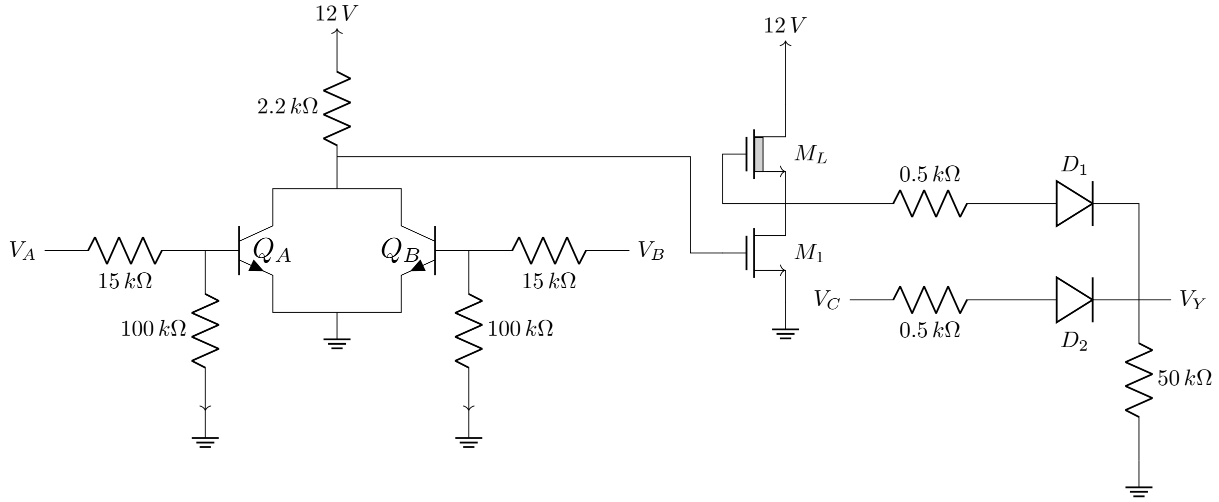

Figure 7

Fill in the following blanks (on the question paper) based on Figure 7—

CO1	(a)	Boolean function of the output: Y = ______________	[1]
CO1	(b)	Minimum value of V_{TN} of M_1 for the inverter to work properly: V_{TN_1} = ______________	[1]
CO1	(c)	If Q_B is in cut-off, D_2 is off & V_Y is high, operating mode of Q_A: ______________	[1]
CO1	(d)	If V_Y is low, operating mode of M_1: ______________	[1]
CO1	(e)	If there were no restrictions on the number of inputs, which of the above gates alone can implement the same logic function? Ans: ______________	[1]
CO1	(f)	If exactly two of the gates were to be used to implement 'Y', which would they be? Ans: ____________________________ How?: ____	[2]
CO1	(g)	If V_A = high, operating mode of M_L: ____________________________	[1]
CO1	(h)	Implement $F = \overline{YB} + A\,\overline{B}$ using exactly one NMOS.	[2]